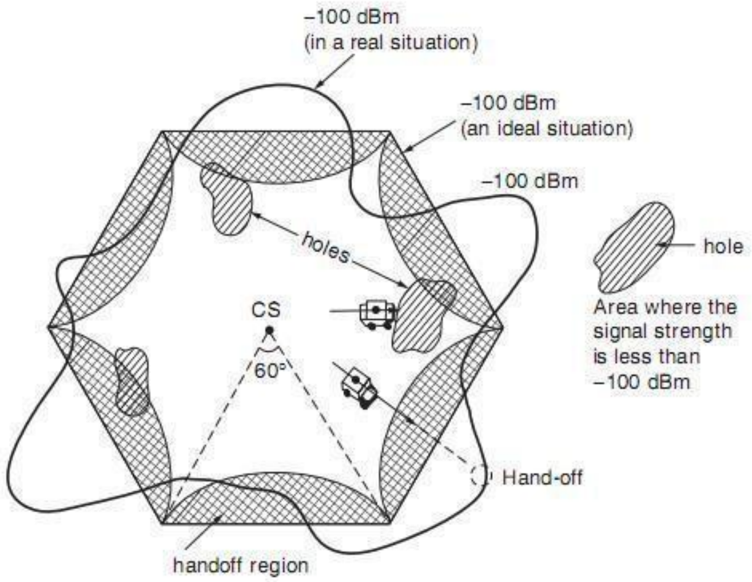
UNIT – III

Handoffs and Dropped Calls

3.1 Why handoffs and types of handoffs

WHY HAND OFF IS NECESSARY

In an analog system, once a call is established, the set-up channel is not used again during the call period. Therefore, handoff is always implemented on the voice channel. In the digital systems, the handoff is carried out through paging or common control channel. The value of implementing handoffs is dependent on the size of the cell. For example, if the radius of the cell is 32 km (20 mi), the area is 3217 km^2(1256 mi^2). After a call is initiated in this area, there is little chance that it will be dropped before the call is terminated as a result of a weak signal at the coverage boundary. Then why bother to implement the handoff feature? Even for a 16-km radius, cell handoff may not be needed. If a call is dropped in a fringe area, the customer simply redials and reconnects the call. Today the size of cells becomes smaller in order to increase capacity. Also people talk longer. The handoffs are very essential. Handoff is needed in two situations where the cell site receives weak signals from the mobile unit: (1) at the cell boundary, say, −100 dBm, which is the level for requesting a handoff in a noise-limited environment; and (2) when the mobile unit is reaching the signal-strength holes (gaps) within the cell site as shown in Fig below.



WHAT ARE THE TWO DECISIONS MAKING PARAMETERS OF HANDOFF

There are two decision-making parameters of handoff: (1) that based on signal strength and (2) that based on carrier-to-interference ratio. The handoff criteria are different for these two types. In type 1, the signal-strength threshold level for handoff is −100 dBm in noise-limited systems and −95 dBm in interference limited systems. In type 2, the value of C/I at the cell boundary for handoff should be at a level, 18 dB for AMPS in order to have toll quality voice. Sometimes, a low value of C/I may be used for capacity reasons.

Type 1: It is easy to implement. The location receiver at each cell site measures all the signal strengths of all receivers at the cell site. However, the received signal strength (RSS) itself includes interference.

$$RSS = C + I$$

where C is the carrier signal power and I is the interference. Suppose that we set up a threshold level for RSS; then, because of the I, which is sometimes very strong, the RSS level is higher and far above the handoff threshold level. In this situation handoff should theoretically take place but does not. Another situation is when I is very low but RSS is also low. In this situation, the voice quality usually is good even though the RSS level is low, but since RSS is low, unnecessary handoff takes place. Therefore, it is an easy but not very accurate method of determining handoffs. Some analog systems use SAT information together with the received signal level to determine handoffs. Some CDMA systems use pilot channel information.

Type 2: Handoffs can be controlled by using the carrier-to-interference ratio C/I.

we can set a level based on C/I ,so C drops as a function of distance but I is dependent on the location. If the handoff is dependent on C/I, and if the C/I drops, it does so in response to increase in (1) propagation distance or (2) interference. In both cases, handoff should take place. In today’s cellular systems, it is hard to measure C/I during a call because of analog modulation. Sometimes we measure the level I before the call is connected, and the level C+I during the call. Thus (C+I)/I can be obtained.

Types of handoffs

A handoff is the process of transferring an active call or data session from one cell to another in a cellular network without interpreting the user. It is a unique feature that allows cellular systems to operate as effectively. There are two types of handoffs

1. Hard handoff 2. Soft handoff.

- **Hard handoff:** The hard handoff is brake before make. In hard handoff the link to the current base station is terminated before or as the user is enters to the new cell's base station. This means that the mobile is not connected at a time to more than one base station.
- **Soft handoff:** The soft handoff is make before brake. Soft handoff is also known as mobile directed handoff because it is directed by the mobile phones. Soft handoff has the ability to select between the instantaneous received signals from different base stations. In soft handoff current channel in the source cell is used for a short time in parallel with the channel in the target cell. In this the connection to the target cell is made before the connection to the source channel is terminated.

Reasons for a Handoff to be conducted:

- To avoid call termination when the mobile user is moving away from the area covered by one cell and entering into the area covered by some another cell.
- When the capacity for connecting new calls of a given cell is full or used up.
- When there is interference in the channels due to the different mobile users using the same channel in different cells.
- When the user behaviors change

3.2 Initiation of handoff

Signal strength is continuously monitored from a reverse voice channel at the cell site. When the signal strength reaches at the desired level of a handoff then the cell site send a request for a handoff on the call to the mobile switching (MSO). A smart decision is made at the cell site that whether the handoff should have taken place earlier or later. If an unnecessary handoff is requested, then the decision was made too early. If a failure handoff occurs, then a decision was made too late.

The some methods are used to make handoffs successful and to reduce/eliminate all needless handoffs. Suppose that −100 dBm is a threshold level of the signal at the cell boundary at which a handoff may be take place. For this situation we must set up signal strength level higher than −100 dBm, suppose it is −100 dBm +Δ dB and when the received signal reaches this level, a handoff request is initiated. If the value of Δ is fixed and large, then the time it takes to lower −100 dBm + Δ to −100 dBm is longer.

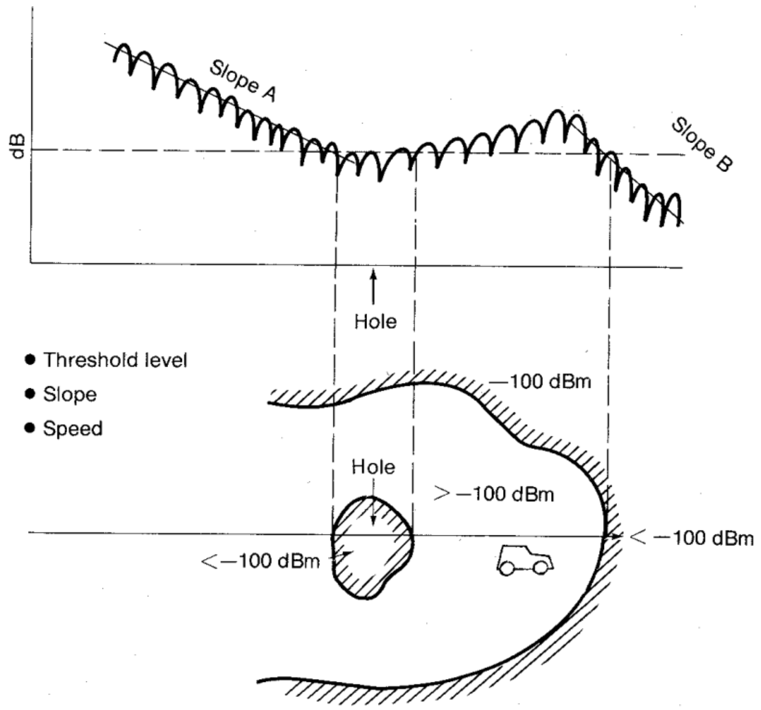


Figure 9.3 Parameters for handling a handoff.

3.3 Delaying a handoff

In many cases, a two-handoff-level algorithm is used. The purpose of creating two request handoff levels is to provide more opportunity for a successful handoff. A handoff could be delayed if no available cell could take the call. A plot of signal strength with two request handoff levels and a threshold level is shown in Fig.3. The plot of average signal strength is recorded on the channel received

Signal strength indicator (RSSI), which is installed at each channel receiver at the cell site. When the signal strength drops below the first handoff level, a handoff request is initiated. If for some reason the mobile unit is in a hole (a weak spot in a cell) or a neighboring cell is busy, the handoff will be requested periodically every 5 s. At the first handoff level, the handoff takes place if the new signal is stronger. However, when the second handoff level is reached, the call will be handed off with no condition. The MSO always handles the handoff call first and the originating calls second. If no neighboring calls are available after the second handoff level is reached, the call continues until the signal strength drops below the threshold level; then the call is dropped. In AMPS

systems if the supervisory audio tone (SAT) is not sent back to the cell site by the mobile unit within 5 s, the cell site turns off the transmitter.

ADVANTAGES OF DELAYED HANDOFF

1. Consider the following example. The mobile units are moving randomly and the terrain contour is uneven. The received signal strength at the mobile unit fluctuates up and down. If the mobile unit is in a hole for less than 5 s (a driven distance of 140 m for 5 s, assuming a vehicle speed of 100 km/h),the delay (in handoff) can even circumvent the need for a handoff. If the neighboring cells are busy, delayed handoff may take place. In principle, when call traffic is heavy, the switching processor is loaded, and thus a lower number of handoffs would help the processor handle call processing more adequately. Of course, it is very likely that after the second handoff level is reached, the call may be dropped with great probability.

2. The other advantage of having a two-handoff-level algorithm is that it makes the handoff occur at the proper location and eliminates possible interference in the system. Figure 3, case I, shows the area where the first-level handoff occurs between cell A and cell B. If we only use the second-level handoff boundary of cell A, the area of handoff is too close to cell B. Figure 3, case II, also shows where the second-level handoff occurs between cell A and cell C. This is because the first-level handoff cannot be implemented.

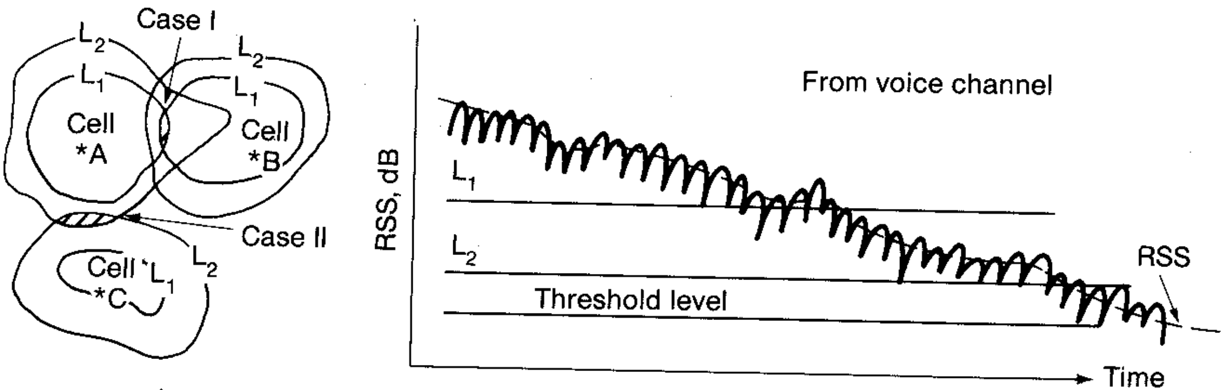


Figure 9.4 A two-level handoff scheme.

3.4 Forced handoffs

A forced handoff is defined as a handoff which is normally would occur but is avoided or prevented from happening, or "a handoff that should not occur but it is forced to happen".

Controlling a Handoff: Controlling of forced handoff can be done by assign a low handoff threshold level a high handoff threshold level. The MSO also can control a handoff as receiving a handoff request from a cell site by making either a handoff earlier or later.

Creating a Handoff

When MSO finds that some cells are having more call load or congested but other cell are not in use or free, in this case the cell site does not request a handoff then the MSO can request call sites to create early handoffs for those congested cells. In such conditions a cell site has to follow the MSO's order and it increases the handoff threshold to push the mobile units at the new boundary and to hand off earlier.

3.6 Mobile assisted handoff and soft handoff

In general handoff mechanism a handoff request is initiated based on measured signal strength mobile signal received by received signal strength indicator (RSSI) at the cell site from the reverse link. The mobile receiver in the digital cellular system is able of monitoring the signal strength of the setup channels of the neighboring cells while serve a call and in a TDMA system, one time slot is used for serving a call and remaining the time slots are used to observe the signal strengths of setup channels. In both systems if the signal strength of its voice channel is weak then the mobile unit can request a handoff and specify which neighboring cell can be assist for handoff to the switching office. The switching office has two set of information: the signal strengths of both forward and reverse setup channels two different neighboring cells or of a neighboring cell. This condition of handoff is known as mobile assisted handoff.

3.6 Cell - site handoff

Cell site handoff scheme can be used in a non-cellular system. The mobile unit has been allotted a frequency in its home cell. When the mobile unit in moving condition and enters a new cell, its frequency does not change but the new cell must tune its frequency as per the frequency of the mobile unit as shown in figure 4.7. In this type of handoff cell sites only need mobile unit the frequency information. This scheme can be implemented only in areas of very low traffic.

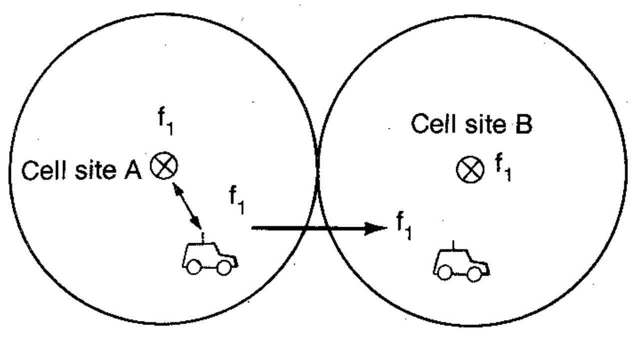


Figure 9.9 Cell-site handoff-only scheme.

3.7 Intersystem handoff

Intersystem handoff occurs when a call initiated in one cellular system which is controlled by one MSO and enters another system which is controlled by another MSO. This condition occurred very less as the maximum call may be terminated before entering in the area of other MSO. Sometimes intersystem handoff can take place i.e. a call handoff can be transferred from one system to a second system so that the call be continued while the mobile unit enters the second system.

The software of MSO must be modified to apply this situation. Consider that car travels on a highway and the driver make a call in system A as shown in figure 4.8. Then the car leaves cell site A of system A and enters cell site B of system B. Here cell sites A is controlled by system A and cell B is controlled by system B. When signal strength of the mobile unit becomes weak in cell site A then MSO A searches for a suitable cell site in its system and but cannot find one. Then MSO A sends the handoff request to MSO B through a dedicated line between MSO A and MSO B, and MSO B makes a complete handoff during the call conversation.

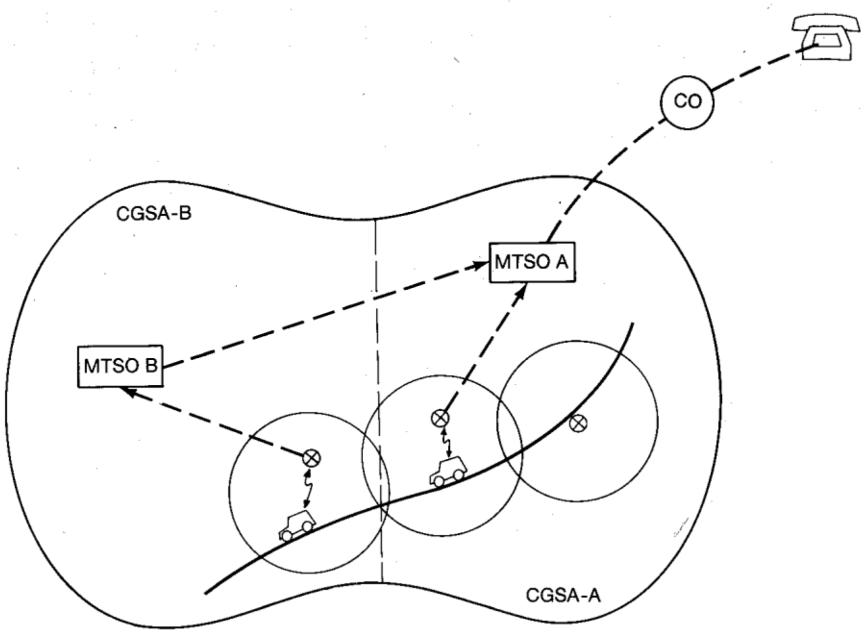


Figure 9.10 Intersystem handoffs.

3.5 Queuing of Handoffs

Queuing of handoffs is more effective than two-threshold-level handoffs. The MTSO will queue the requests of handoff calls instead of rejecting them if the new cell sites are busy. A queuing scheme becomes effective only when the requests for handoffs arrive at the MTSO in batches or bundles. If handoff requests arrive at the MTSO uniformly, then the queuing scheme is not needed. Before showing the equations, let us define the parameters as follows.

$1/\mu$	average calling time in seconds, including *new calls* and *handoff calls* in each cell
λ_1	arrival rate (λ_1 calls per second) for originating calls
λ_2	arrival rate (λ_2 handoff calls per second) for handoff calls
M_1	size of queue for originating calls
N	number of voice channels
a	$(\lambda_1 + \lambda_2)/\mu$
b_1	λ_1/μ
b_2	λ_2/μ

The following analysis can be used to see the improvement. We are analyzing three cases.[8]

1. *No queuing on either the originating calls or the handoff calls.* The blocking for either a originating call or a handoff call is

$$B_o = \frac{a^N}{N!} P(0) \qquad (9.5\text{-}1)$$

where

$$P(0) = \left(\sum_{n=0}^{N} \frac{a^N}{n!}\right)^{-1} \tag{9.5-2}$$

2. *Queuing the originating calls but not the handoff calls.* The blocking probability for originating calls is

$$B_{oq} = \left(\frac{b_1}{N}\right)^{M_1} P_q(0) \tag{9.5-3}$$

where

$$P_q(0) = \left[N! \sum_{n=0}^{N-1} \frac{a^{n-N}}{n!} + \frac{1 - (b_1/N)^{M_1+1}}{1 - (b_1/N)}\right]^{-1} \tag{9.5-4}$$

The blocking probability for handoff calls is

$$B_{oh} = \frac{1 - (b_1/N)^{M_1+1}}{1 - (b_1/N)} P_q(0) \tag{9.5-5}$$

3. *Queuing the handoff calls but not the originating calls.* The blocking

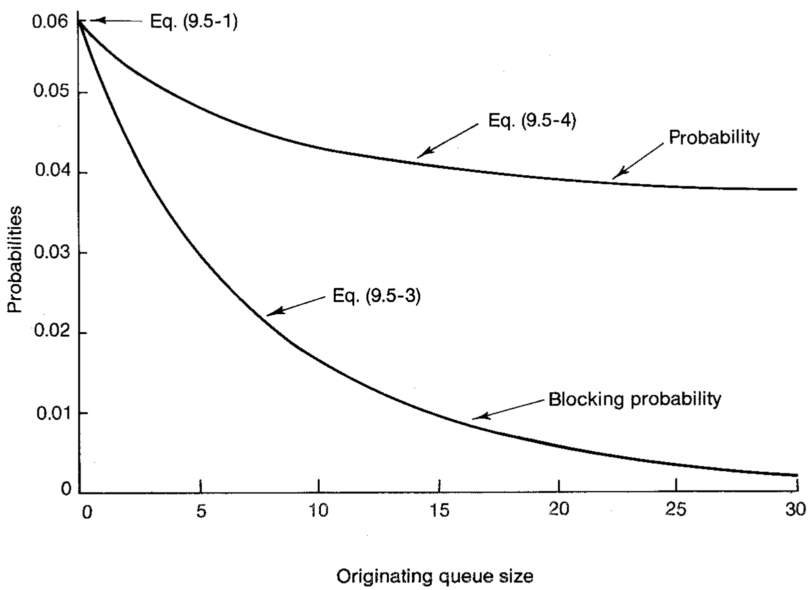


Figure 9.5 Probability and blocking probability graph showing blocking probability for originating calls queuing for originating calls ($N = 70$).

probability for handoff calls is

$$B_{hq} = \left(\frac{b_2}{N}\right)^{M_2} P_q(0) \tag{9.5-6}$$

where $P_q(0)$ is as shown in Eq. (9.5-4). The blocking probability for originating calls is

$$B_{ho} = \frac{1 - (b_2/N)^{M_2+1}}{1 - (b_2/N)} P_q(0) \tag{9.5-7}$$

Example 9.1 The following parameters are given. The number of channels at the cell site $N = 70$. The call holding time is 101 s = 0.028 h. The number of originating calls attempted per hour is expressed as $\lambda_1 = 2270$. The number of handoff calls attempted per hour is expressed as $\lambda_2 = 80$. Then

$$A = \frac{\lambda + \lambda_2}{\mu} = (2270 + 80)\ 0.028 \doteq 65.80$$

$$b_1 = \frac{\lambda_1}{\mu} = 2270 \times 0.028 = 63.60$$

$$b_2 = \frac{\lambda_2}{\mu} = 2.24$$

Given these parameters, Eqs. (9.5-1), (9.5-3), (9.5-5), (9.5-6), and (9.5-7) have been plotted in Figs. 9.5, 9.6, 9.7, and 9.8 respectively.

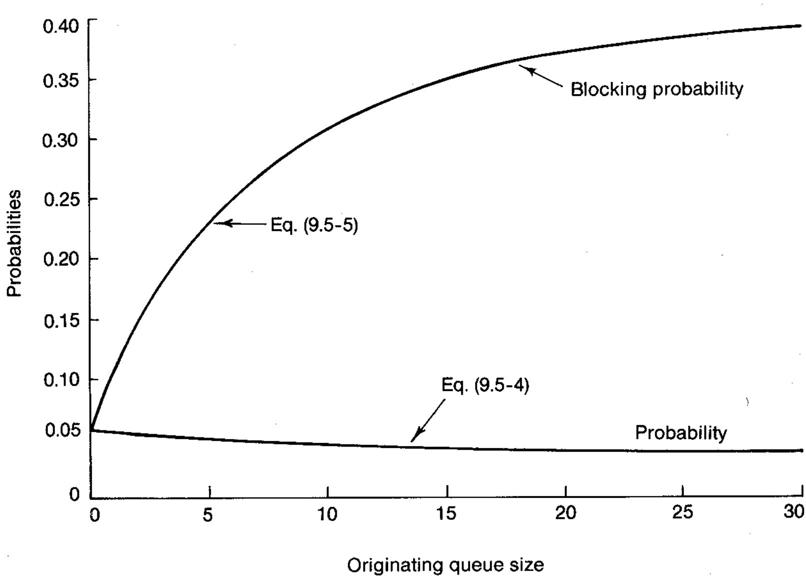


Figure 9.6 Probability and blocking probability graph showing blocking probability for handoff calls queuing for originating calls ($N = 70$).

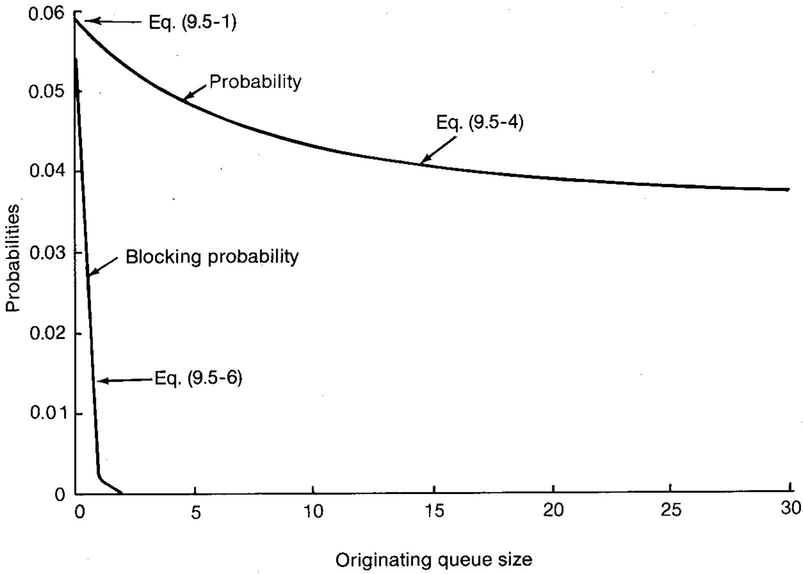


Figure 9.7 Probability and blocking probability graph showing blocking probability for handoff calls (queuing for handoff calls) ($N = 70$).

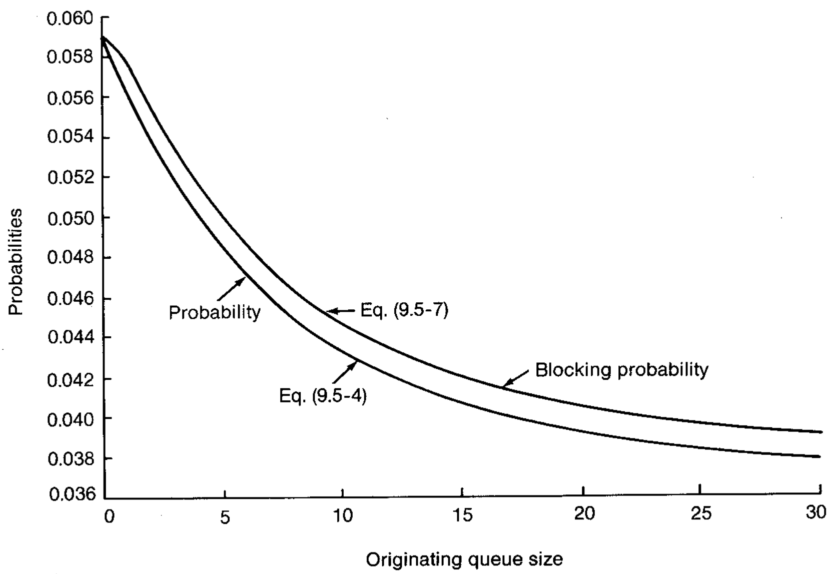


Figure 9.8 Probability and blocking probability graph showing blocking probability for originating calls (queuing for handoff calls) ($N = 70$).

We have seen (Figs. 9.5 and 9.6) with queuing of originating calls only, the probability of blocking is reduced. However, queuing of originating calls results in increased blocking probability on handoff calls, and this is a drawback. With queuing of handoff calls only, blocking probability is reduced from 5.9 to 0.1 percent by using one queue space (see Fig. 9.7). Therefore it is very worthwhile to implement a simple queue (one space) for handoff calls. Adding queues in handoff calls does not affect the blocking probability of originating calls in this particular example (see Fig. 9.8). However, we should always be aware that queuing for the handoff is more important than queuing for those initiating calls on assigned voice channels because call drops upset customers more than call blockings.

3.6 Power-Difference Handoffs

A better algorithm is based on the power difference (Δ) of a mobile signal received by two cell sites, home and handoff. Δ can be positive or negative. The handoff occurs depending on a preset value of Δ.

$$\Delta = \text{the mobile signal measured at the candidate handoff site} - \text{the mobile signal measured at the home site} \tag{9.6-1}$$

For example, the following cases can occur.

$\Delta > 3$ dB	request a handoff
$1 \text{ dB} < \Delta < 3$ dB	prepare a handoff
$-3 \text{ dB} < \Delta < 0$ dB	monitoring the signal strength
$\Delta < -3$ dB	no handoff

Those numbers can be changed to fit the switch processor capacity. This algorithm is not based on the received signal strength level, but on a relative (power difference) measurement. Therefore, when this algorithm is used, all the call handoffs for different vehicles can occur at the same general location in spite of different mobile antenna gains or heights.

9.10 Introduction to Dropped Call Rate

The definition of dropped call rate. The definition of a dropped call is after the call is established but before it is properly terminated. The definition of "the call is established" means that the call is setup completely by the setup channel. If there is a possibility of a call drop due to no available voice channels, this is counted as a blocked call not a dropped call.

If there is a possibility that a call will drop due to the poor signal of the assigned voice channel, this is considered a dropped call. This case can happen when the mobile or portable units are at a standstill and the radio carrier is changed from a strong setup channel to a weak voice channel due to the selective frequency fading phenomenon.

The perception of dropped call rate by the subscribers can be higher due to:

1. The subscriber unit not functioning properly (needs repair).
2. The user operating the portable unit in a vehicle (misused).
3. The user not knowing how to get the best reception from a portable unit (needs education).

Consideration of dropped calls. In principle, dropped call rate can be set very low if we do not need to maintain the voice quality. The dropped call rate and the specified voice quality level are inversely proportional. In designing a commercial system, the specified voice quality level is given relating to how much C/I (or C/N) the speech coder can tolerate. By maintaining a certain voice quality level, the dropped call rate can be calculated by taking the following factors into consideration:

1. Provide signal coverage based on the percentage (say 90%) that all the received signal will be above a given signal level.
2. Maintain the specified co-channel and adjacent channel interference levels in each cell during a busy hour, i.e., the worst interference case.
3. Since the performance of the call dropped rate is calculated as possible call dropping in every stage from the radio link to the PSTN connection, the response time of the handoff in the network will be a factor when the cell becomes small, the response time for a handoff request has to be shorter in order to reduce the call dropped rate.
4. The signaling of the handoff and the MAHO algorithm will also impact the call dropped rate.
5. The relationship among the voice quality, system capacity and call dropped rate can be expressed through a common parameter C/I.

Relationship among capacity, voice quality, dropped call rate. Radio Capacity m is expressed as follows:

$$m = \frac{B_T/B_c}{\sqrt{\frac{2}{3}\,(C/I)_S}} \tag{9.10-1}$$

where B_T/B_c is the total number of voice channels. B_T/B_c is a given number, and $(C/I)_S$ is a required C/I for designing a system. The above equation is obtained based on six co-channel interferers which occur in busy traffic, i.e., a worst case. In an interference limited system, the adjacent channel interference has only a secondary effect. The derivation of Eq. (9.10-1) will be expressed in Chap. 13. Eq. (9.10-1) can be changed to the following form:

$$(C/I)_S = \frac{3}{2}\left(\frac{B_T/B_c}{m}\right)^2 = \frac{3}{2}\left(\frac{B_T}{B_c}\right)^2 \cdot \frac{1}{m^2} \tag{9.10-2}$$

Since the $(C/I)_S$ is a required C/I for designing a system, the voice quality is based on the $(C/I)_S$. When the specified $(C/I)_S$ is reduced, the radio capacity is increased. When the measured (C/I) is less than the specified $(C/I)_S$, both poor voice quality and dropped calls can occur.

Coverage of 90% equal-strength contour. The coverage in cellular cells always uses the coverage of 90% equal-strength contour. The prediction tool (Lee Model) described in Chap. 4 is used to predict the equal-strength contour at level C with 50% time and 50% area in a cell. For example, let $C = -102$ dBm, which is 18 dB above the ambient noise -120 dBm. If $C = -102$ dBm is 50% equal-strength contour, then increase the level to $C + 10$ dB contour which can be calculated from the following equation:

$$P(x' < A) = \int_{-\infty}^{A} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[\frac{(y' - \overline{m})^2}{2\sigma^2}\right] dy'$$

$$= P\left(x < \frac{A - \overline{m}}{\sigma}\right) \qquad (9.10\text{-}3)$$

Eq. (9.10-3) is the cumulative distribution function where A is the desired signal level and $\overline{m}$ is the mean level. σ is long-term fading due to terrain contour. If $A = C + 10 = -92$ dB and $\sigma = 8$ dB:

$$S = P\left(x < \frac{-92 - (-102)}{8}\right) = P\left(x < \frac{10}{8}\right) = 0.9082 \qquad (9.10\text{-}4)$$

Eq. (9.10-4) can also be interpreted as being at a -92 dBm contour, the signal above the level of -102 dBm is 90.8%. Of course, the level of -102 dBm is determined to be 18 dB above -120 dBm which is the ambient noise level. The $(C/N)_S$ of 18 dB is the required level for getting a voice quality.